

Authoring M11/USDM Protocols for Structured Review and Reuse

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Abstract

Clinical trial protocols remain central to study design, regulatory review, ethics review, site conduct, data collection, statistical analysis, registry disclosure, and submission preparation. In many current workflows, the protocol is authored primarily as a document, and operational content is later reinterpreted for data capture, monitoring, statistical planning, registry reporting, and submission-related activities. This repeated reinterpretation creates avoidable variation between the approved protocol and downstream artifacts.

This paper builds on prior work evaluating structured protocol content and downstream reuse, including a prior white paper and presentations delivered at the 2026 DIA China Annual Meeting and the 2026 CDISC Europe Interchange. That work described the use case and initial results. This paper focuses on what is needed to move from demonstration to implementation: authoring protocols aligned with ICH M11 and the Unified Study Definitions Model (USDM) from the start, building validation into authoring, defining structured Schedule of Activities (SOA) expectations, preserving traceability, clarifying regulatory submission and review expectations, and aligning the roles of sponsors, standards organizations, technology providers, and national regulatory authorities (NRAs).

The principal lesson is that structured protocol implementation should not be framed as conversion of a completed DOCX or PDF into data. The more reliable approach is structured authoring, where protocol content is created, validated, reviewed, rendered, amended, and reused as controlled structured content. The structured SOA should be an early test case because it contains visits, assessments, timing, and conditional activities that downstream workflows repeatedly reinterpret today.

USDM API 4.0.0 provides computable study-definition structures that can support M11 implementation, but M11 and USDM serve related rather than interchangeable functions. Reliable implementation will require documented implementation rules for M11-to-USDM representation, interim structured SOA expectations, validation and rendering evidence, and regional implementation requirements or technical specifications where needed. Where legally permitted, certain technical requirements may need to be binding so that structured protocols can be exchanged and reviewed consistently across NRAs.

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1. Introduction

The clinical trial protocol is both a scientific document and an operational specification. It defines objectives, endpoints, eligibility criteria, interventions, visits, activities, assessments, timing, statistical considerations, safety monitoring, and governance expectations. These elements are used by different stakeholders for different purposes, but they originate from a common study design.

In practice, the approved protocol often reaches downstream systems as a DOCX or PDF document. Teams then interpret that document and create derivative artifacts such as electronic case report forms, edit checks, monitoring plans, statistical programming specifications, safety review plans, registry entries, trial master file artifacts, and Study Data Tabulation Model (SDTM) mapping specifications [1]. The process is familiar and workable, but it is not computationally stable. Different teams may interpret the same protocol element differently, especially when timing rules, conditional assessments, footnotes, optional procedures, or amendment history are complex.

The technical question is not whether protocol documents should disappear. Human-readable protocols remain necessary for review, communication, archival purposes, and regulatory operations. The question is whether the protocol document should remain the only primary form of protocol content. A structured protocol representation can preserve human-readable outputs while also providing computable design metadata for validation, reuse, and traceability.

2. Background: Prior Use Case and Results

This paper builds on prior work rather than replacing it. The earlier white paper and the China and Milan presentations described the use case and presented results from evaluating structured protocol content and downstream reuse [2-4]. The use case established the practical starting point: this content can support more than document rendering, but the workflow must be governed, validated, and connected to downstream processes to be useful in regulated clinical research.

The prior work also identified a central implementation issue. Reuse is strongest when structure is introduced during authoring rather than recovered after final document creation. Post-hoc extraction from DOCX or PDF can be useful for legacy protocols and transitional workflows, but it remains a recovery step. It must infer data elements, relationships, terminology, timing, and conditions from content designed primarily for human reading.

The results of the use case therefore create a next-step question: what is required to make structured protocol authoring reliable, usable for regulatory review, and scalable? This paper focuses on lessons learned, the final M11 Technical Specification, USDM, structured SOA implementation, validation built into authoring, downstream traceability, and regulatory implementation requirements.

3. Purpose and Scope

Building on prior use case work that examined structured protocol content and downstream reuse, this paper focuses on what is needed to move from demonstration to reliable implementation now that the ICH M11 package is final and USDM API 4.0.0 is available as a current implementation specification.

In this paper, USDM refers to the study definition model, while USDM API 4.0.0 refers to the specific API specification reviewed for implementation and exchange considerations [5, 6]. The broader Digital Data

Flow initiative provides additional context for structured study-definition content and downstream reuse across the clinical study lifecycle [7].

The analysis addresses how structured protocol content should be authored, validated, rendered, reviewed, and reused across downstream processes. It gives particular attention to the structured Schedule of Activities, M11-to-USDM alignment, built-in validation, traceability, regulatory submission expectations, and the implementation rules or technical specifications needed to support broader implementation.

The next phase will require action by multiple groups. Sponsors need to test whether M11/USDM protocols can be authored from the start in realistic protocol-development workflows. Standards organizations need to clarify implementation rules, especially for M11-to-USDM representation and the structured SOA. Technology providers need to support authoring, validation, rendering, exchange, review, and downstream reuse without forcing sponsors back into document-first workflows. National regulatory authorities (NRAs) need to define expectations for structured protocol submission, review, exchange, and archiving so implementation does not fragment across authorities.

4. Regulatory and Standards Context

U.S. Food and Drug Administration (FDA) announced the availability of the final guidance for industry entitled M11 Clinical Electronic Structured Harmonised Protocol (CeSHarP) in the Federal Register on May 22, 2026. The notice states that the guidance was prepared under ICH and includes three documents: a guidance, a template, and a technical specification document. It describes the guidance as providing the rationale and recommendations for implementing a harmonized digital clinical trial protocol, the template as providing standardized content structure and formatting, and the technical specification as containing harmonized terminologies and standardized data fields for electronic exchange of clinical protocol information [8].

The M11 Technical Specification states that its purpose is to serve as a technical representation of the ICH M11 protocol template. It describes protocol information using table elements such as term, data type, heading/data/value designation, definition, user guidance, conformance, cardinality, relationship to the table of contents, values, business rules, and repeating or reuse rules. It also points to ICH M11 terminology in NCI Thesaurus and ICH object identifiers available through ICH ESTRI [9].

USDM API 4.0.0 is an OpenAPI specification that defines data structures and API interfaces for representing and exchanging computable study definitions. The API specification identifies itself as Digital Data Flow USDM API version 4.0.0 and includes model content relevant to study design, study conduct, timing, activities, documents, amendments, identifiers, organizations, devices, and terminology-related information [6].

5. Key Lesson 1: Structure Should Begin During Authoring

The first lesson from the use case is that structured protocol implementation should not be reduced to extraction from a final document. Extracting structured information from a completed DOCX or PDF can support legacy conversion, document review, and transitional data capture. It does not, however, provide the same control as creating and governing structured protocol content during authoring.

In a structured authoring workflow, the study definition is created and maintained as structured content from the start of protocol development. The human-readable protocol remains an output, but it is

generated from controlled structured content rather than serving as the only source from which structure is later inferred. This changes where structure is introduced and how it is controlled.

Structured authoring does not remove the need for scientific judgment, medical writing expertise, operational review, or regulatory assessment. It supports those functions by making protocol elements, relationships, terminology, timing, amendments, and rendering behavior available for validation and reuse. That is the practical shift: from repeated reinterpretation of a document to controlled reuse of governed protocol metadata.

6. Key Lesson 2: M11-to-USDM Implementation

The second lesson is that M11 and USDM serve related but different functions. The M11 Technical Specification is a technical representation of the ICH M11 protocol template and describes protocol information in terms of headings, data elements, values, definitions, conformance, cardinality, template hierarchy, controlled terminology, business rules, and repeat/reuse expectations [9]. USDM API 4.0.0 defines data structures and API interfaces for representing and exchanging computable study definition information [6].

The practical implementation question is how protocol content aligned with ICH M11 should be represented, validated, rendered, exchanged, and reviewed using USDM-supported structured content. This will require documented implementation rules for representing M11 Technical Specification elements, generating the human-readable protocol from structured content, applying terminology and identifiers, and producing and retaining validation evidence.

Several areas illustrate why implementation rules will be needed. Eligibility criteria, SOA timing, and amendment history may each require decisions about how protocol text, timing windows, conditions, footnotes, and version changes are represented in structured content and rendered for review. Without documented rules, sponsors and technology providers may make these decisions differently, increasing the risk of inconsistent implementation described later in this paper.

These implementation rules should be developed through practical testing and should inform future versions of USDM, related FHIR implementation guide activities, and regional technical specifications. The objective is to determine where current M11 and USDM artifacts are sufficient, where additional guidance or model evolution may be needed, and how that content can be exchanged and reviewed consistently across sponsors, technology providers, standards organizations, and national regulatory authorities.

7. Key Lesson 3: The Structured SOA Is the Practical Starting Point

The third lesson is that the SOA is the most practical first use case for structured protocol implementation. It sits at the point where protocol design becomes operational: what must be done, when it must be done, for whom it applies, and how the results are used. Because the SOA drives data collection, endpoint assessment, monitoring, analysis planning, registry disclosure, and submission-related workflows, its timing rules and conditional logic need to be represented as structured content rather than reconstructed from a table.

A structured SOA should represent more than a visual table. It should capture the activities, visits, timing windows, planned assessments, and conditional activities that define how the study is conducted. It should also preserve the information needed for review and conduct, including procedure groupings, relevant controlled terminology, links to objectives and endpoints where applicable, labels, and footnotes.

A common set of implementation rules for the structured SOA will be needed to avoid fragmentation. These rules should explain how SOA content is represented, rendered, validated, and exchanged. They should specify the required and optional SOA elements, the controlled terms to be used, how timing and conditional logic are represented, how amendments are handled, how the SOA is displayed for review, and what conformance checks apply.

8. Key Lesson 4: Validation Must Be Built into Authoring

The fourth lesson is that protocol validation should be built into the structured authoring workflow. It should not be treated as a final check after a document or exchange package has already been generated. In a USDM-supported authoring process, validation should begin when structured content is created and continue as the protocol is reviewed, rendered, and amended.

The purpose is not simply to determine whether a final structured exchange package is technically valid. A technically valid package may still be scientifically incomplete, inconsistent with the rendered protocol, or unsuitable for regulatory review. Conversely, a scientifically complete protocol may fail exchange or conformance checks if required structure, terminology, relationships, cardinality, or metadata are missing or inconsistent.

The authoring environment should therefore check the protocol as it is being developed. These checks should confirm that required content is present, terminology is valid, relationships are complete, timing logic is consistent, amendments are tracked, and the rendered protocol matches the structured content.

Downstream use still needs confirmation, but that is not the same as authoring validation. Once governed structured content is approved, downstream teams should confirm that it has been interpreted correctly for electronic data capture (EDC) configuration, source data extraction (SDE), Statistical Analysis Plan (SAP) inputs, registry content, SDTM mapping, and submission package preparation.

Validation results should be retained as evidence. The structured protocol record should include, or link to, validation reports that identify the rules applied, terminology versions, execution dates, results, warnings, errors, reviewer dispositions, and change history. This evidence supports internal governance, amendment control, and regulatory review if structured protocol submission becomes expected.

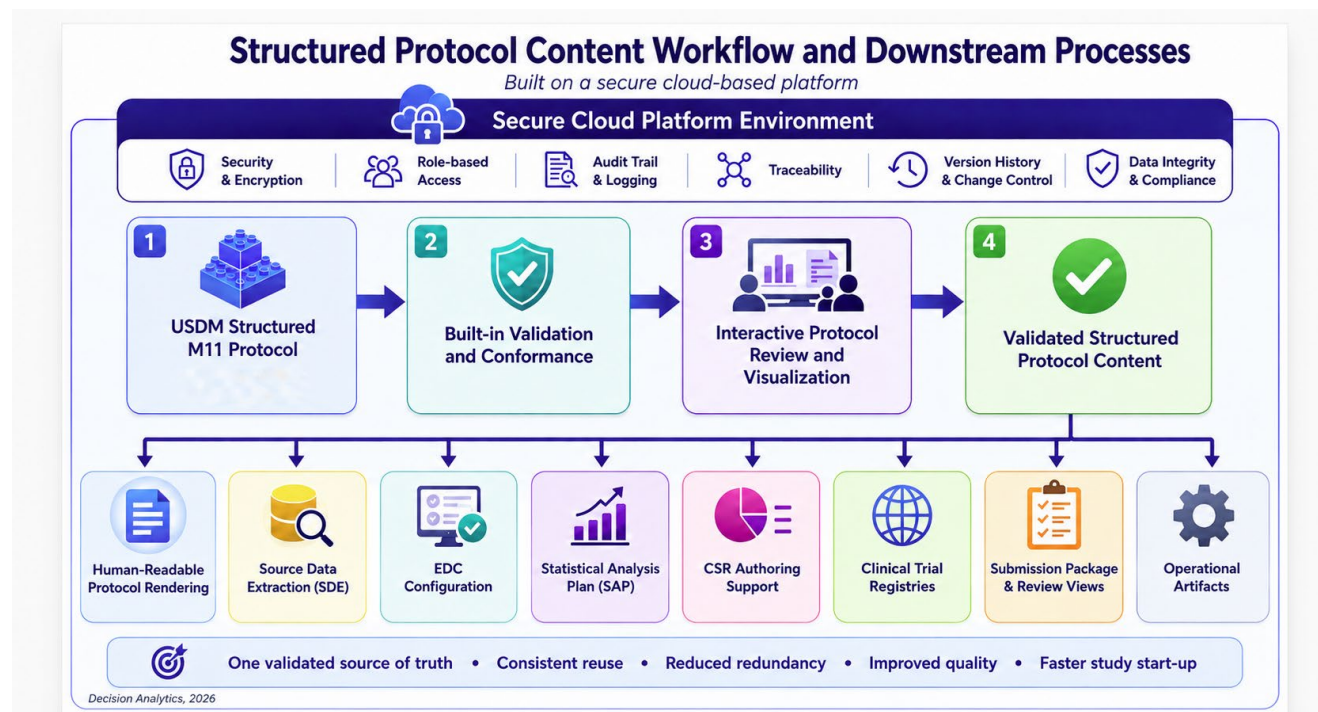
9. Key Lesson 5: Downstream Reuse Requires Traceability and Governance

The fifth lesson is that downstream reuse is not simply exporting protocol data. Reuse is valuable only if structured protocol content remains traceable to the approved protocol, the applicable version, the review decisions, and the downstream artifact that uses it. Without that traceability, structured content can become another disconnected copy of the protocol rather than a governed source for reuse.

Governance is equally important. Governed protocol content should be reused through controlled workflows that preserve role-based access, version history, validation results, reviewer actions, and audit trails. Figure 1 summarizes the intended workflow: protocol content is authored, validated, reviewed, and

reused within a secure cloud-based platform where security, traceability, version control, and data integrity are built into the environment.

Figure 1



The prior use case showed the clearest measured downstream benefit in source data extraction. In that workflow, source documents were uploaded directly to the SDE platform, redacted in-platform, processed for extraction, and reviewed with a retained electronic source link. Compared with manual transcription of source data into EDC, the SDE workflow reduced manual re-entry and preserved the relationship among the protocol requirement, redacted source evidence, extracted value, and monitor review action.

The results supported this conclusion. In the sponsor use case presented in the prior work, the SDE workflow met or exceeded predefined targets for extraction accuracy, optical character recognition and natural language processing flag accuracy, personally identifiable information redaction accuracy, provenance integrity, and transcription reduction. Measured clinical data coordinator plus monitor workflow time decreased from 84 minutes per patient for manual source-data transcription and review to 39 minutes per patient using the source-linked SDE workflow.

The same traceability principle applies to other downstream uses. Structured protocol content can provide a governed reference for EDC configuration, SAP development, clinical study report (CSR) authoring, registry disclosure, SDTM mapping, and submission package preparation. These outputs still require their own review, but they should remain connected to the protocol requirements, version history, validation evidence, and review decisions from which they were derived.

For SDTM-related workflows, structured protocol content provides context for planned observations. It does not create SDTM datasets by itself, but it can support dataset planning and mapping by making planned activities, assessment timing, endpoint relationships, visits, and controlled terminology easier to inspect and reuse. The value is not only efficiency; it is the ability to preserve a traceable path from protocol requirements to collected or extracted data and downstream submission-related outputs.

10. Regulatory Expectations for Structured Protocols

Finalization of ICH M11 creates the need for clear expectations on how structured protocols will be submitted, processed, reviewed, archived, and exchanged. The common foundation should be the ICH M11 protocol template and technical specification. Regional implementation requirements may still be needed, but they should build on that common foundation so structured protocols can be used consistently within and across national regulatory authorities.

Where the legal framework permits, certain technical requirements may need to be made binding to ensure consistent structured protocol exchange and review. At minimum, agencies should clarify how structured protocol content relates to the human-readable protocol, how conformance to the ICH M11 Technical Specification will be evaluated, and what validation evidence should accompany a structured protocol submission.

A common M11-based exchange approach is also important for regulatory reliance and internal agency review. If multiple authorities are reviewing the same clinical development program, they should be able to inspect and compare the same structured protocol content using compatible tools. Within an agency, a repository of computable protocols could support comparison across studies and therapeutic areas, including review of objectives, endpoints, eligibility criteria, SOA content, amendment history, and study design features. These uses should inform metadata, terminology, validation, and review-tool expectations.

The structured SOA requires separate attention. NRAs should not prescribe how sponsors conduct a trial, but they can define expectations for how submitted SOA content is structured, exchanged, rendered, and validated as part of an M11-based structured protocol. If the ICH M11 artifacts become binding or are adopted into regional requirements, the format and structure of the submitted SOA may also need to be specified. This could be addressed through regional technical specifications or by a standards development organization, such as CDISC, which may be well positioned to define a standard structured SOA model and conformance rules.

11. Next Phase Evaluation Priorities

The next phase should build directly from the prior use case. That work showed that structured protocol content, once created, can support validation, visualization, source-linked extraction, provenance, and downstream SDTM planning readiness. However, the prior work began with protocol content engineered into M11-compliant structured form from document-based sources. The next phase should test whether sponsors can author and use M11/USDM protocols from the start.

The next phase should also recognize two implementation horizons. The longer-term goal is sponsor authoring of M11/USDM protocols from the start, with validation built into authoring and the human-readable protocol rendered from the same controlled structured content. The nearer-term need is a set of practical tools that help sponsors validate, render, compare, and refine structured protocol content

during the transition from document-first workflows. These tools should not become the end state, but they can help sponsors, technology providers, standards organizations, and regulators learn what implementation rules are needed.

This phased approach is consistent with implementation-focused commentary emphasizing USDM as a foundation for downstream use cases and the need to build and consume study definitions as a continuum [10, 11].

The real requirement is broader than any single authoring tool. The workflow needs authoring, validation, rendering, exchange, review, and downstream reuse to work together. The next-phase priorities should therefore test not only whether structured content can be created, but whether it can retain meaning, traceability, and review utility as it moves across systems and organizations.

Priority 1: Authoring M11/USDM Protocols from the Start

Sponsors should evaluate whether their authoring tools and internal processes can produce protocol content that conforms to the ICH M11 template and technical specification. Validation should be built into authoring so that required content, terminology, structure, rendering, versioning, and amendment information can be checked before a final protocol package is generated. The human-readable protocol should be generated from the same governed structured content, not maintained as a separate source.

This priority should build on prior M11-to-USDM representation work. The next phase should document, at an implementation level, how ICH M11 Technical Specification elements are represented: directly in USDM, through extensions or other implementation approaches, through rendering, through submission or review metadata, or as unresolved items requiring further work. This mapping is necessary to move from general alignment to implementable conformance.

Priority 2: Structured SOA and Downstream Reuse

A structured SOA should be the principal downstream test case. In the absence of a formal standard SOA model, pilots should use documented interim SOA implementation rules. These rules should define how SOA activities, visits, timing, conditions, amendments, rendering, and review checks are represented for the pilot. It should not be presented as a final industry standard; its purpose is to generate evidence that can inform additional ICH implementation work, or regional technical specifications.

This priority should test whether upfront sponsor-authored M11/USDM structured protocol content can drive source data extraction and other downstream uses. The SDE pilot should evaluate whether, in the implemented workflow, protocol requirements remain traceable to redacted source evidence, extracted values, monitor or sponsor review action, and SDTM-related outputs. Additional pilots should assess whether this content can support preparation of protocol-derived SAP content and selected registry disclosure content for ClinicalTrials.gov and other national or international registries, recognizing that registry-specific data models, required fields, and submission rules will still require mapping, review, and completion.

Priority 3: Regulatory Review, Reliance, and Implementation Expectations

The next phase should include regulatory review use cases. Where national regulatory authorities participate, the focus should be on whether structured protocol content improves review. Regulators should be able to inspect, search, compare, and visualize protocol elements such as objectives, endpoints, eligibility criteria, SOA activities, amendments, and study design features.

This priority should also evaluate how structured protocols could support regulatory reliance. If multiple authorities are reviewing the same clinical development program, they should be able to review compatible structured protocol content using common tools. The findings should distinguish what can be implemented using the current M11 and USDM artifacts from what still needs additional technical direction, including M11 implementation rules, a structured SOA model or standard, and regional regulatory expectations.

Together, these three priorities should produce implementation evidence, not just demonstrations. They should test whether structured protocol content can be authored upstream, reused downstream, and reviewed by regulators in a way that is consistent, traceable, and scalable.

12. Risks and Controls

The most important risk is inconsistent implementation. If national regulatory authorities, sponsors, vendors, and standards implementations apply M11 structured protocol requirements differently, structured protocol content will not be reusable at scale. This would limit the downstream value that structured protocol content is intended to support.

This risk is especially important for the structured SOA. If each tool or sponsor represents activities, visits, timing, conditional logic, amendments, terminology, and rendering differently, the industry will recreate the same interpretation problem that exists today with document-first workflows. Structured content would exist, but its operational and regulatory value would be limited.

The primary control is a common M11-based implementation approach. Regional requirements may still be needed, but they should build on the ICH M11 template and technical specification rather than create separate implementation paths. Common expectations should address structured protocol exchange, validation evidence, rendering, amendment handling, terminology use, structured SOA representation, and review-tool capabilities.

Next phase pilots should test whether M11/USDM protocol content authored by sponsors can be exchanged, validated, reviewed, and reused without losing meaning or traceability. These pilots should identify where current M11 and USDM artifacts are sufficient, where interim implementation rules are needed, and where standards-development organizations or NRAs should provide additional technical direction.

13. Conclusion

The prior use case showed that structured protocol content can support downstream reuse, but it also showed that the next phase must move upstream. The focus should shift from converting completed protocol documents into structured data to testing whether sponsors can author M11/USDM protocols from the start. Those pilots should build on the prior M11-to-USDM representation work and evaluate interim structured SOA expectations, built-in validation, rendering, traceability, and downstream reuse.

The results should inform standards development work and regional implementation requirements. Where legally permitted, certain technical requirements may need to be binding so that structured protocols can be exchanged and reviewed consistently across national regulatory authorities. The objective is not simply to digitize the protocol, but to make protocol content reusable, traceable, and reviewable across the clinical development lifecycle.

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